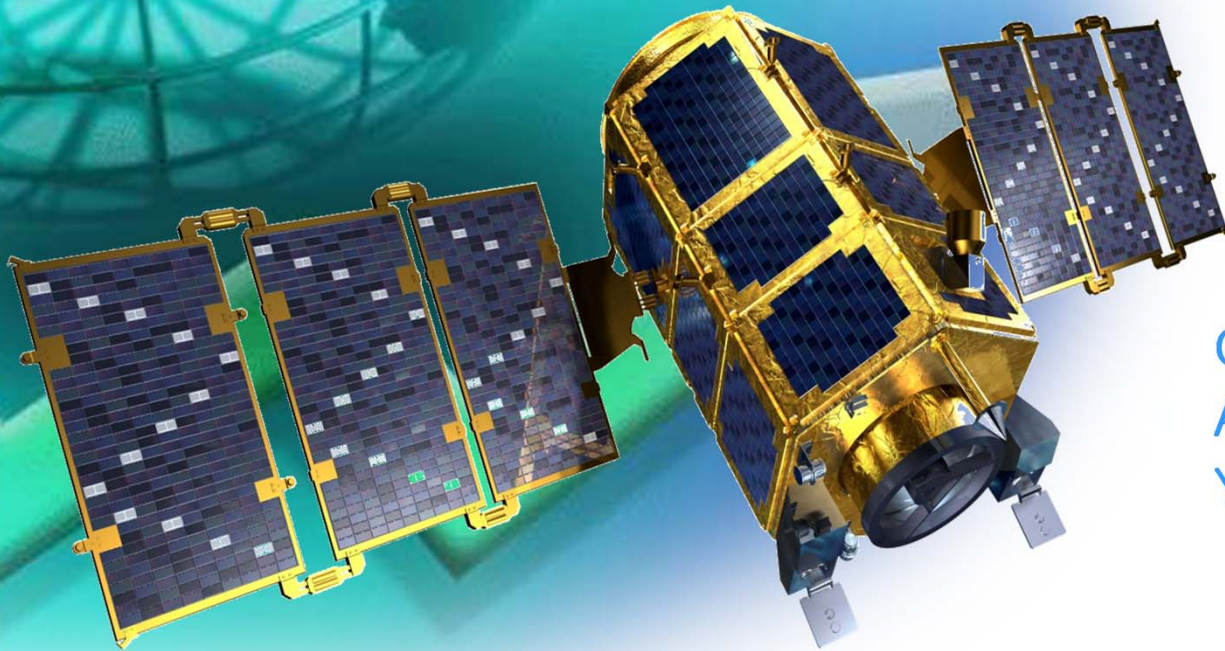
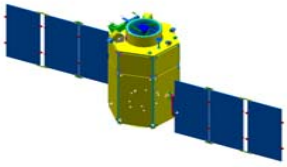


Fall 2011

Spacecraft Systems



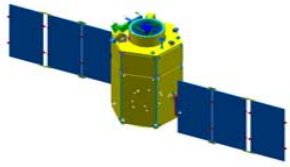
Chandeok Park
Astrodynamics & Control Lab.
Yonsei University



Chapter 11

SPACECRAFT SUBSYSTEMS

- 11.1 Attitude Determination & Control Subsystem
- 11.2 Telemetry, Tracking & Command Subsystem
- 11.3 Command & Data Handling Subsystem
- 11.4 (Electrical) Power Subsystem**
- 11.5 Thermal Control Subsystem
- 11.6 Structures & Mechanism Subsystem
- 11.7 Guidance, Navigation & Control Subsystem



11.4 Electrical Power Subsystem

◆ EPS

- Provides/Stores/Distributes/Controls S/C electrical power
- Average Power at EOL (end of life) determines the size of Power Source
- Power at BOL (beginning of life) = $(2\sim3) \times (\text{Power at EOL})$

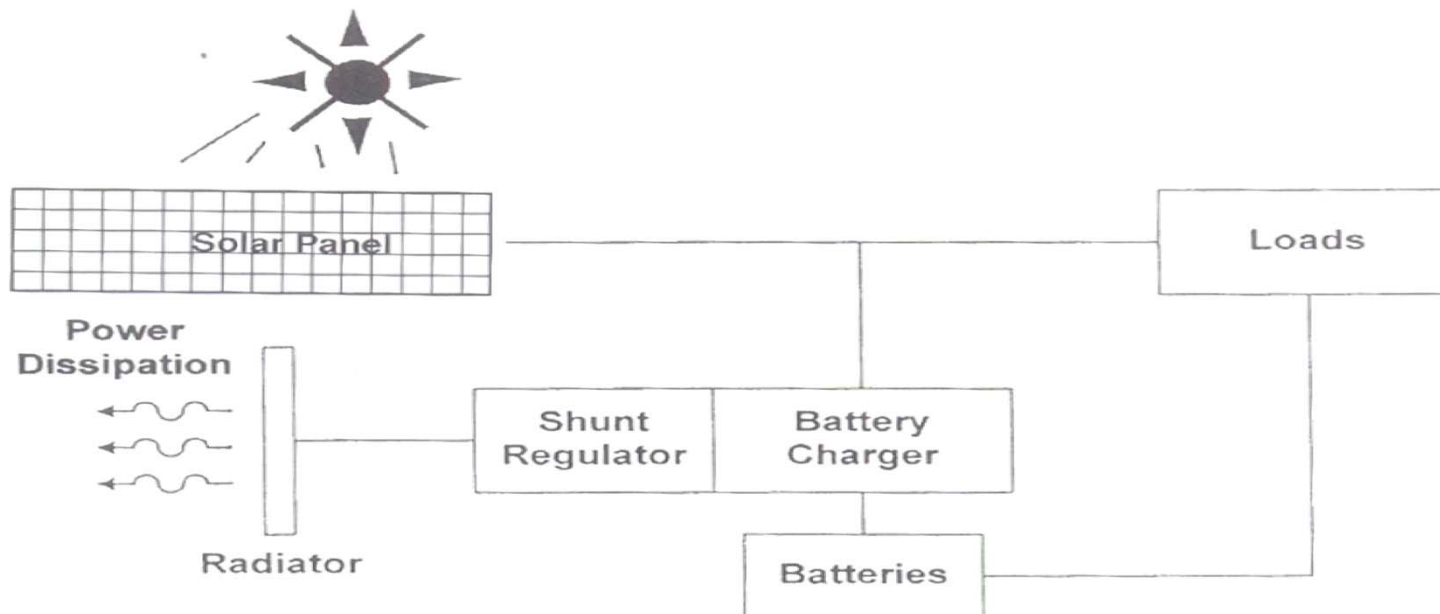
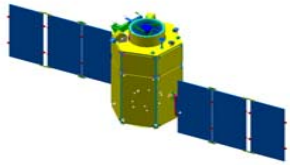


Fig. 6.2 Typical solar panel-battery power system (direct energy transfer architecture).



11.4 EPS

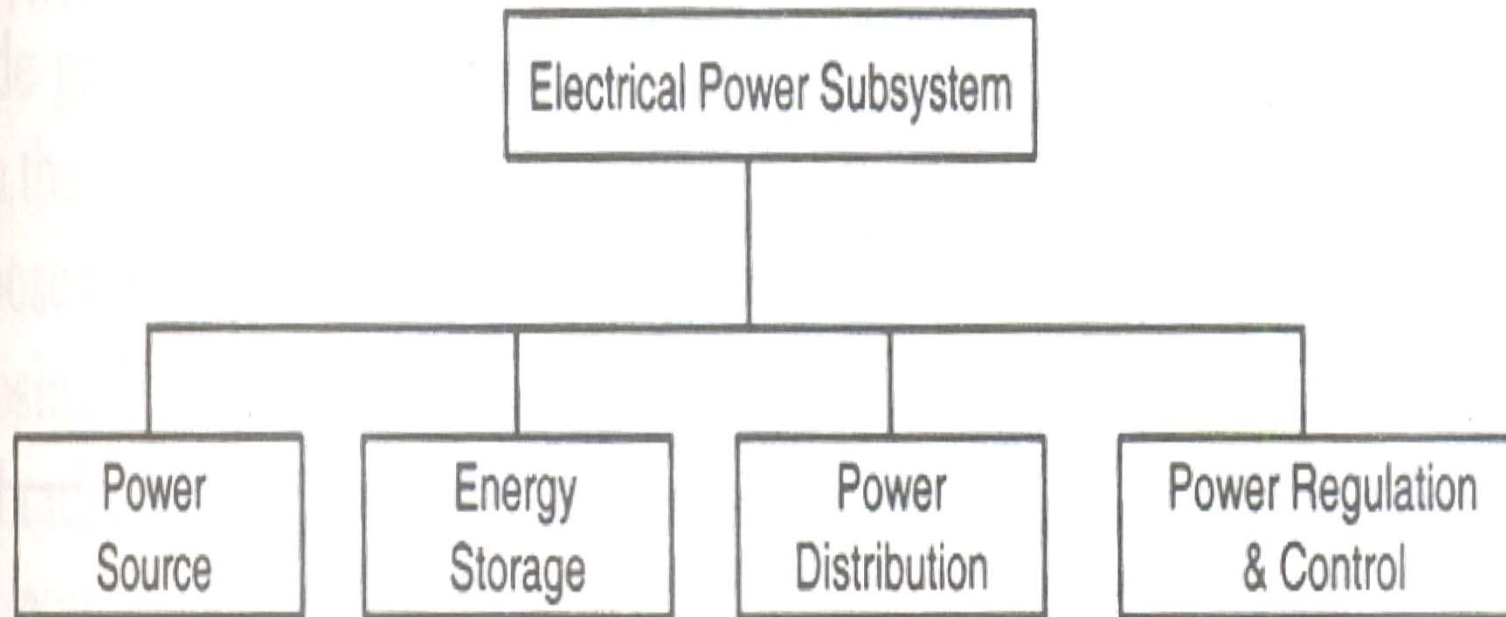
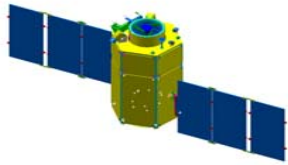


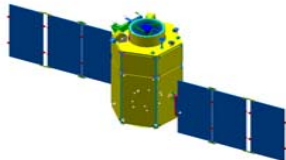
Fig. 11-8. **Functional Breakdown for the Spacecraft's Power Subsystem.** We start with these four functions and must determine requirements for the hardware, software, and interfaces for each.



11.4 EPS

TABLE 11-30. Typical Top-Level Power Subsystem Functions. Each of these functions consists of subfunctions with a myriad design characteristics which we must develop to meet mission requirements.

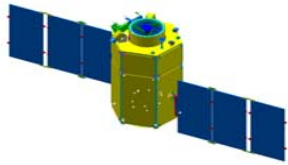
- Supply a continuous source of electrical power to spacecraft loads during the mission life.
- Control and distribute electrical power to the spacecraft.
- Support power requirements for average and peak electrical load.
- Provide converters for ac and regulated dc power buses, if required.
- Provide command and telemetry capability for EPS health and status, as well as control by ground station or an autonomous system.
- Protect the spacecraft payload against failures within the EPS.
- Suppress transient bus voltages and protect against bus faults.
- Provide ability to fire ordnance, if required.



11.4 EPS

TABLE 11-31. The Preliminary Design Process for the Power Subsystem. All of these design steps must link back to mission requirements to satisfy the owner and users. Note that derived requirements may impact previous design decisions and force designers to iterate the design process.

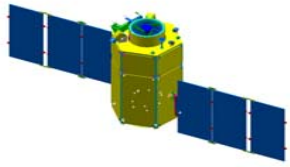
Step	Information Required	Derived Requirements	References
1. Identify Requirements	Top-level requirements, mission type (LEO, GEO), spacecraft configuration, mission life, payload definition	Design requirements, spacecraft electrical power profile (average and peak)	Secs. 10.1, 10.2
2. Select and Size Power Source	Mission type, spacecraft configuration, average load requirements for electrical power	EOL power requirement, type of solar cell, mass and area of solar array, solar array configuration (2-axis tracking panel, body-mounted)	Secs. 10.1, 10.2 Table 10-9 Sec. 11.4.1 Table 11-34
3. Select and Size Energy Storage	Mission orbital parameters, average and peak load requirements for electrical power	Eclipse and load-leveling energy storage requirement (battery capacity requirement), battery mass and volume, battery type	Sec. 11.4.2 Tables 11-3, 11-4, 11-38, 11-39, 11-40 Fig. 11-11
4. Identify Power Regulation and Control	Power-source selection, mission life, requirements for regulating mission load, and thermal-control requirements	Peak-power tracker or direct-energy-transfer system, thermal-control requirements, bus-voltage quality, power control algorithms	Sec. 11.4.4



11.4 EPS

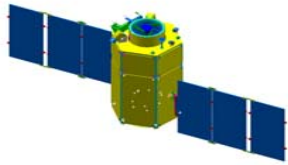
TABLE 11-32. Effects of System-Level Parameters on the Power Subsystem. Most aspects of the mission affect the power subsystem because so many other subsystems require specific power attributes.

Parameter	Effects on Design
<i>Average Electrical Power Requirement</i>	Sizes the power-generation system (e.g., number of solar cells, primary battery size) and possibly the energy-storage system given the eclipse period and depth of discharge
<i>Peak Electrical Power Required</i>	Sizes the energy-storage system (e.g., number of batteries, capacitor bank size) and the power-processing and distribution equipment
<i>Mission Life</i>	Longer mission life (> 7 yr) implies extra redundancy design, independent battery charging, larger capacity batteries, and larger arrays
<i>Orbital Parameters</i>	Defines incident solar energy, eclipse/Sun periods, and radiation environment
<i>Spacecraft Configuration</i>	Spinner typically implies body-mounted solar cells; 3-axis stabilized typically implies body-fixed and deployable solar panels



◆ 11.4.1 Power Sources

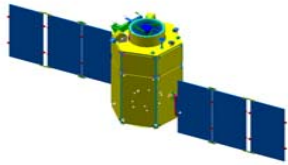
- Photovoltaic Solar Cells
 - Convert incident solar radiation directly to electrical energy
 - Most common power source for Earth-orbiting S/C
- Static Power
 - Uses a heat source (plutonium 238 or uranium 235) for direct thermal-to-electric conversion
- Dynamic Power
 - Uses a heat source (plutonium 238, enriched uranium) to provide electrical power using Brayton, Stirling, or Rankine cycles.
- Fuel Cells
 - Convert the chemical energy of an oxidation reaction to electricity



◆ 11.4.1 Power Sources (cont'd)

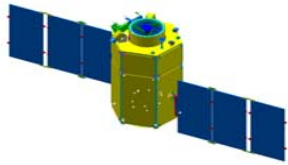
TABLE 11-33. Matrix for Comparing Most Common Spacecraft Power Sources. We may use different factors to select the correct power source but specific power and specific cost are used extensively.

EPS Design Parameters	Solar Photovoltaic	Solar Thermal Dynamic	Radio-isotope	Nuclear Reactor	Fuel Cell
Power Range (kW)	0.2–300	5–300	0.2–10	5–300	0.2–50
Specific Power (W/kg)	25–200	9–15	5–20	2–40	275
Specific Cost (\$/W)	800–3,000	1,000–2,000	16K–200K	400K–700K	Insufficient Data
Hardness – Natural Radiation – Nuclear Threat – Laser Threat – Pellets	Low–Medium Medium Medium Low	High High High Medium	Very high Very high Very high Very high	Very high Very high Very high Very high	High High High Medium



◆ 11.4.1 Power Sources (cont'd)

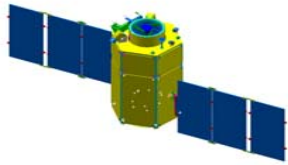
Stability and Maneuverability	Low	Medium	High	High	High
Low-orbit Drag	High	High	Low	Medium (due to radiator)	Low
Degradation Over Life	Medium	Medium	Low	Low	Low
Storage Required for Solar Eclipse	Yes	Yes	No	No	No
Sensitivity to Sun Angle	Medium	High	None	None	None
Sensitivity to Spacecraft Shadowing	Low (with bypass diodes)	High	None	None	None
Obstruction of Spacecraft Viewing	High	High	Low	Medium (due to radiator)	None
Fuel Availability	Unlimited	Unlimited	Very low	Very low	Medium
Safety Analysis Reporting	Minimal	Minimal	Routine	Extensive	Routine
IR Signature	Low	Medium	Medium	High	Medium
Principal Applications	Earth-orbiting spacecraft	Interplanetary, Earth-orbiting spacecraft	Inter-planetary	Inter-planetary	Inter-planetary



◆ 11.4.1 Power Sources (cont'd)

TABLE 11-34. Solar Array Design Process. In the FireSat example column, I_d represents inherent degradation, θ is the Sun incidence angle, L_d is life degradation, and X_e and X_d represent the efficiencies of the power distribution paths. The material following the table further explains these quantities.

Step	Reference	FireSat Example
1. Determine requirements and constraints for power subsystem solar array design • Average power required during daylight and eclipse • Orbit altitude and eclipse duration • Design lifetime	Input parameter, Secs. 10.1, 10.2 Input parameter, end papers Chaps. 2, 3	110 W during daylight and eclipse 700 km 35.3 min 5 yr
2. Calculate amount of power that must be produced by the solar arrays, P_{sa}	Step 1 Eq. 5-5, end papers (Orbit period – T_e) Eq. 11-5	$P_e = P_d = 110$ W $T_e = 35.3$ min $T_d = 63.5$ min Assume a peak power tracking regulation scheme with $X_e = 0.6$ and $X_d = 0.8$ $P_{sa} = 239.4$ W

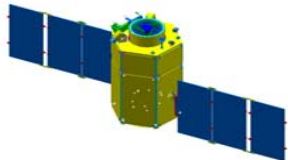


◆ 11.4.1 Power Sources (cont'd)

3. Select type of solar cell and estimate power output, P_o , with the Sun normal to the surface of the cells	*Si: $P_o = 0.148 \times 1,367 \text{ W/m}^2 = 202 \text{ W/m}^2$ *GaAs: $P_o = 0.185 \times 1,367 \text{ W/m}^2 = 253 \text{ W/m}^2$ *Multijunction: $P_o = 0.22 \times 1,367 \text{ W/m}^2 = 301 \text{ W/m}^2$	Si solar cells $P_o = 202 \text{ W/m}^2$
4. Determine the beginning-of-life (BOL) power production capability, P_{BOL} , per unit area of the array	Table 11-35 Eq. 5-7 Eq. 11-6	$I_d = 0.77$ $\theta = 23.5 \text{ deg (worst case)}$ $P_{BOL} = 143 \text{ W/m}^2$
5. Determine the end-of-life (EOL) power production capability, P_{EOL} , for the solar array	Performance degradation Si: 3.75% per yr, GaAs: 2.75% per yr, Multijunction: 0.5% per yr Eq. 11-7 Eq. 11-8	Performance degradation is 3.75% per year $L_d = 0.826$ for 5 yr mission $P_{EOL} = 118.1 \text{ W/m}^2$
6. Estimate the solar array area, A_{sa} , required to produce the necessary power, P_{sa} , based on P_{EOL} an alternate approach	Eq. 11-9 Eq. 10-12†	$A_{sa} = 2.0 \text{ m}^2$ $A_{sa} = 2.5 \text{ m}^2$
7. Estimate the mass of the solar array	Eq. 10-13†	$M_a = 9.6 \text{ kg}$
8. Document assumptions		

* Typical demonstrated efficiencies for Si, GaAs, and multijunction solar cells are 14.8%, 18.5%, and 22%, respectively.

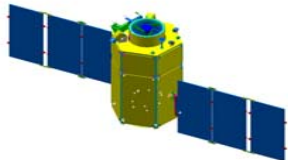
† Use P_{sa} in these equations.



◆ 11.4.1 Power Sources (cont'd)

TABLE 11-35. Performance Comparison for Photovoltaic Solar Cells. Note that the stated efficiencies are for single solar cells, not solar arrays.

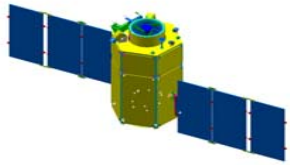
Cell Type	Silicon	Thin Sheet Amorphous Si	Gallium Arsenide	Indium Phosphide	Multijunction GaInP/GaAs
Planar cell theoretical efficiency	20.8%	12.0%	23.5%	22.8%	25.8%
Achieved efficiency:					
Production	14.8%	5.0%	18.5%	18%	22.0%
Best laboratory	20.8%	10%	21.8%	19.9%	25.7%
Equivalent time in geosynchronous orbit for 15% degradation					
– 1 MeV electrons	10 yr	10 yr	33 yr	155 yr	33 yr
– 10 MeV protons	4 yr	4 yr	6 yr	89 yr	6 yr



◆ 11.4.1 Power Sources (cont'd)

TABLE 11-36. Elements of Inherent Solar Array Degradation. Although individual solar cells may have adequate efficiency, after we manufacture the solar array, these elements cause some degradation in the cumulative efficiency by the amounts indicated.

Elements of Inherent Degradation	Nominal	Range
<i>Design and Assembly</i>	0.85	0.77–0.90
<i>Temperature of Array</i>	0.85	0.80–0.98
<i>Shadowing of Cells</i>	1.00	0.80–1.00
<i>Inherent Degradation, I_d</i>	0.77	0.49–0.88



◆ 11.4.1 Power Sources (cont'd)

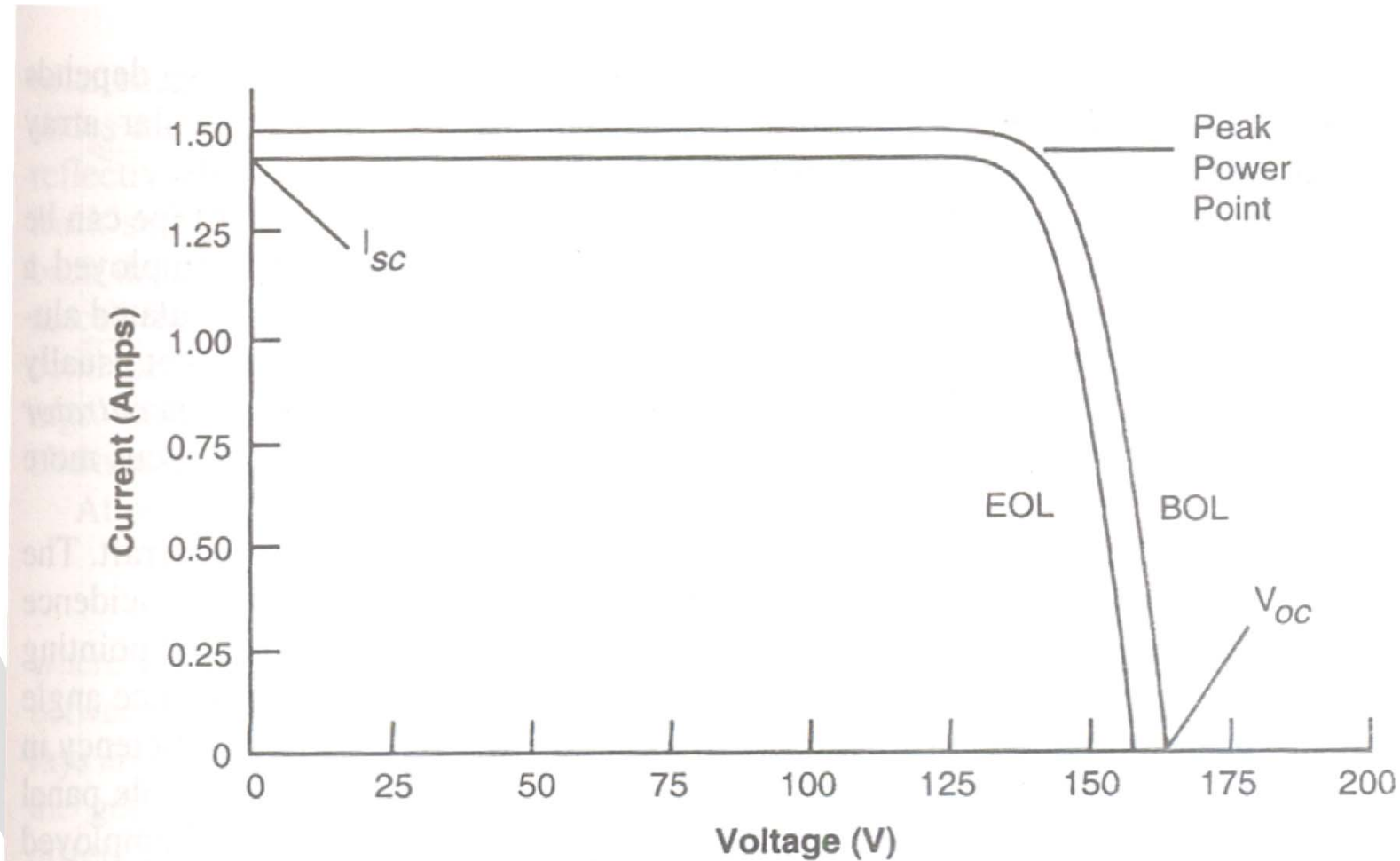
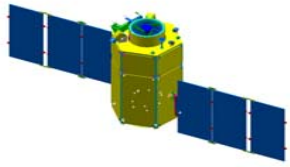
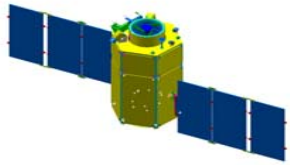


Fig. 11-9. I-V Plot for a Planar Array. The power available is simply the area under the curve.



◆ 11.4.2 Energy Storage

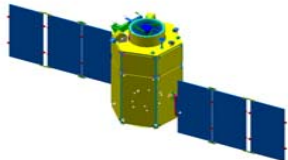
- Any S/C that uses photovoltaic or solar thermal dynamics as a power source requires a system to store energy for peak-power demands and eclipse periods
 - Energy storage typically occurs in a battery
 - Battery consists of individual cells connected
- The amount of energy stored with the battery is ampere-hour capacity or watt-hour capacity
 - $(\text{Watt-Hour Capacity}) = (\text{Ampere-Hour}) * (\text{Operating Voltage})$



◆ 11.4.2 Energy Storage (cont'd)

TABLE 11-37. Issues in Designing the Energy Storage Capability. Energy storage usually means large batteries and we must consider all their characteristics when designing this subsystem.

<i>Physical</i>	Size, weight, configuration, operating position, static and dynamic environments
<i>Electrical</i>	Voltage, current loading, duty cycles, number of duty cycles, activation time and storage time, and limits on depth-of-discharge
<i>Programmatic</i>	Cost, shelf and cycle life, mission, reliability, maintainability, and produceability



◆ 11.4.2 Energy Storage (cont'd)

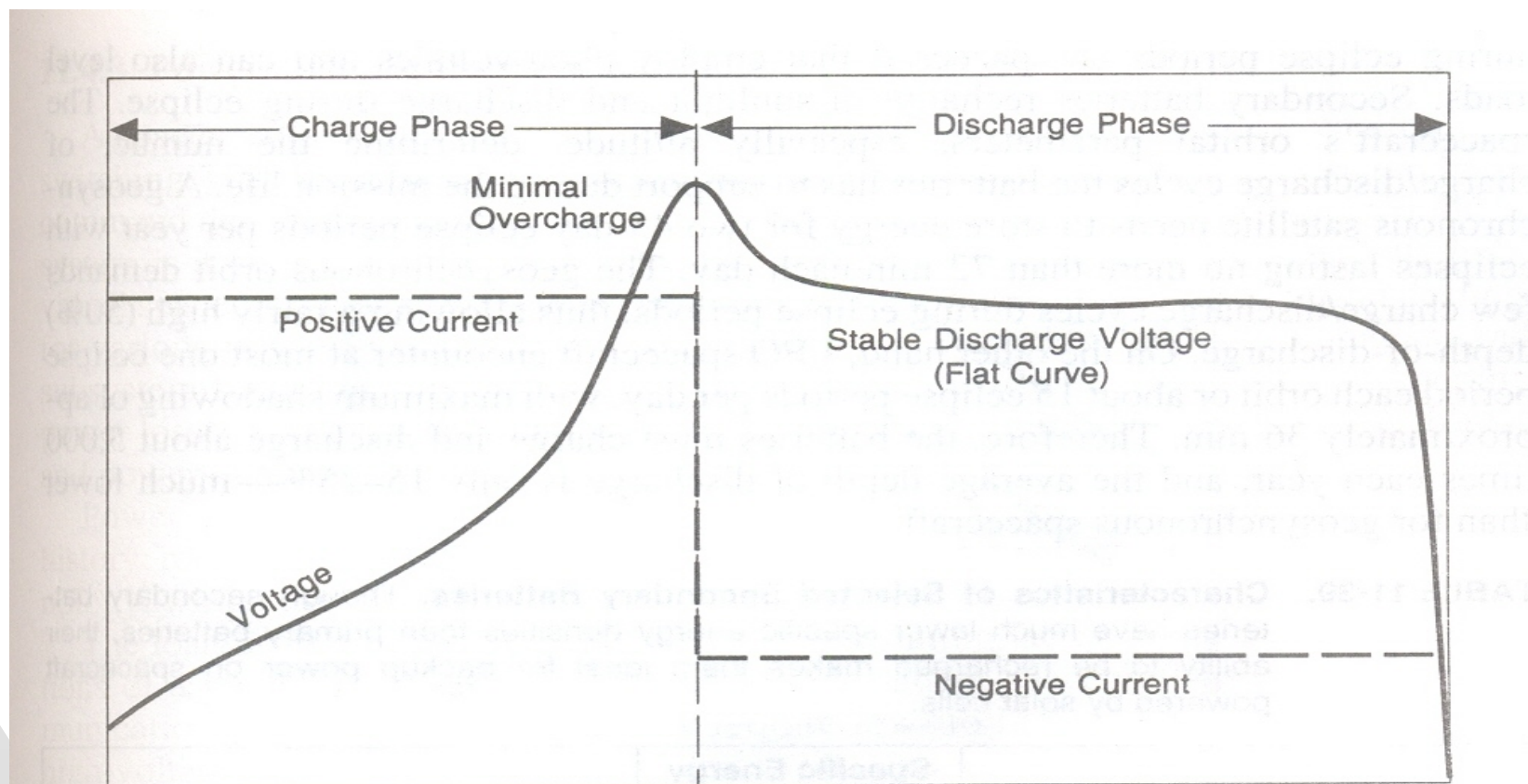
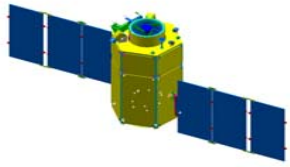


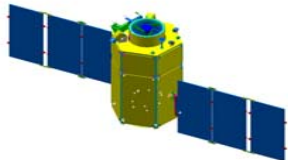
Fig. 11-10. Profile of Charge/Discharge Voltages for Batteries. Secondary batteries may cycle through this type of profile hundreds or thousands of times during their mission life. At the left edge, the voltage is low because the spacecraft just came out of eclipse where it used battery power. During the charge phase, there is positive current from the power regulator, so the battery voltage rises. In the discharge phase (in eclipse again), there is a negative current, so the battery voltage decreases.



◆ 11.4.2 Energy Storage (cont'd)

- Primary Battery Cells
 - Convert chemical energy into electrical energy
 - Cannot reverse this conversion, so they cannot be recharged

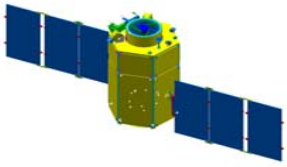
- Secondary Battery for Energy Storage
 - Can convert chemical/electrical energy into electrical/chemical energy during discharge/charge
 - Provides power during eclipse periods



◆ 11.4.2 Energy Storage (cont'd)

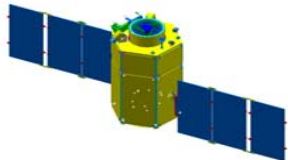
TABLE 11-39. Characteristics of Selected Secondary Batteries. Though secondary batteries have much lower specific energy densities than primary batteries, their ability to be recharged makes them ideal for backup power on spacecraft powered by solar cells.

Secondary Battery Couple	Specific Energy Density (W·hr/kg)	Status
Nickel-Cadmium	25 – 30	Space-qualified, extensive database
Nickel-Hydrogen (individual pressure vessel design)	35 – 43	Space-qualified, good database
Nickel-Hydrogen (common pressure vessel design)	40 – 56	Space-qualified for GEO and planetary
Nickel-Hydrogen (single pressure vessel design)	43 – 57	Space-qualified
Lithium-Ion (LiSO_2 , LiCF , LiSOCl_2)	70 – 110	Under development
Sodium-Sulfur	140 – 210	Under development



◆ 11.4.2 Energy Storage (cont'd)

- LEO S/C
 - One Eclipse Period/Orbit (Maximum Shadowing): ~36 minutes
 - 15 Eclipses/Day
 - ~5000 Charge and Discharge/Year
- Average Depth of Discharge: Only 15~ 25%
 - Depth of Discharge (DOD) is simply the percent of total battery capacity removed during a discharge period



◆ 11.4.2 Energy Storage (cont'd)

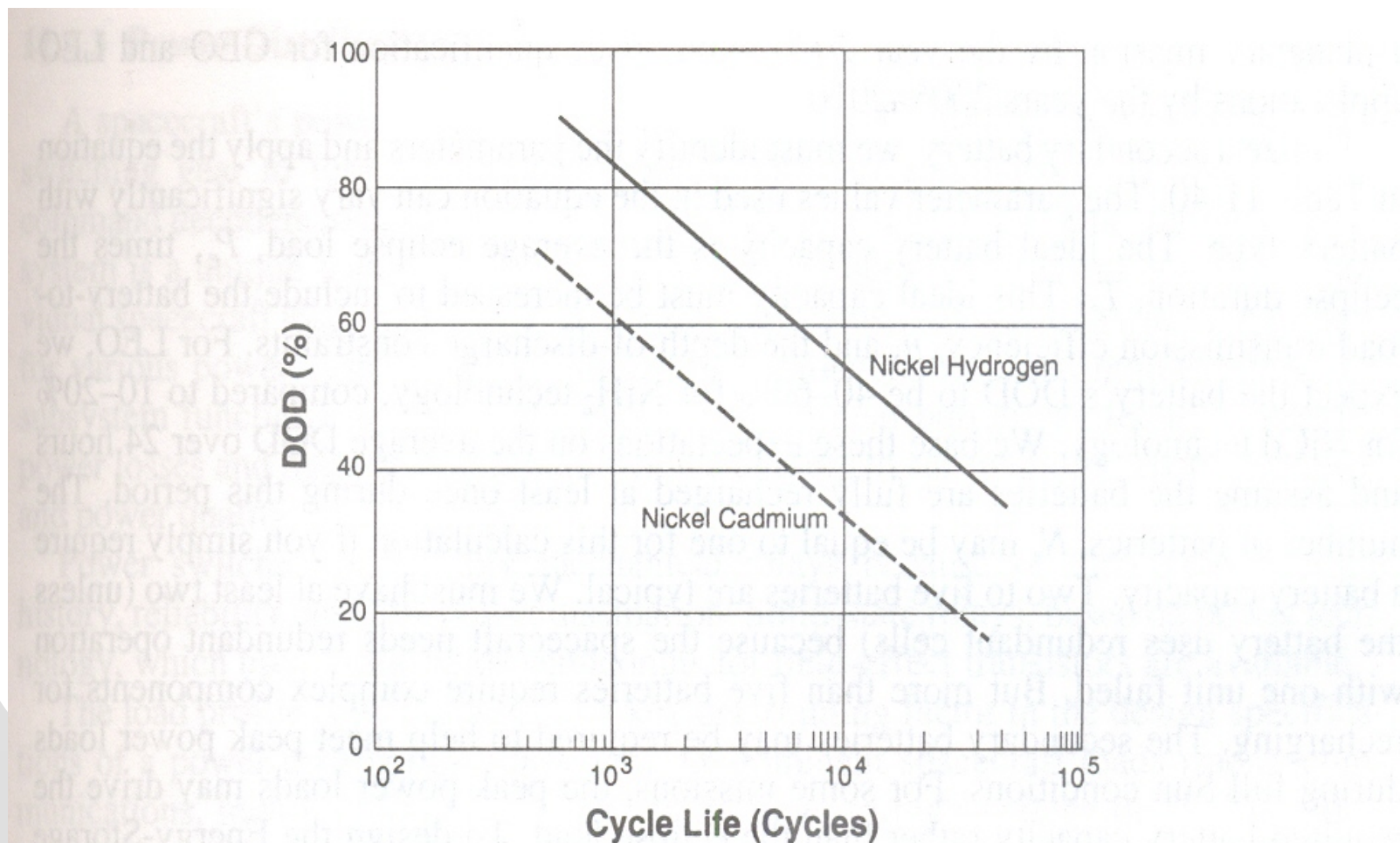
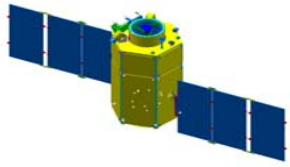
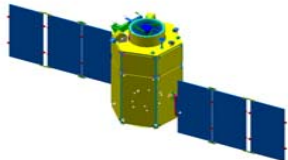


Fig. 11-11. Depth-of-Discharge vs. Cycle Life for Secondary Batteries. Increased cycle life reduces the amount of energy available from the batteries during each cycle—DOD decreases with cycle life.



◆ 11.4.2 Energy Storage (cont'd)

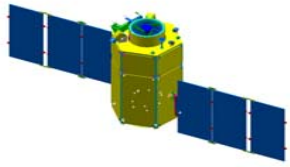
- NiCd battery
 - Still a common secondary energy storage system for many aerospace applications
 - 28V DC, 22~23 series cell, 5~100 Amp-Hour
- Sizing a secondary battery requires us to identify the parameters and apply equations in Table 11-40
- Battery Capacity
 - $C_r = P_e T_e / (DOD) N n$ (W-Hr) or $C_r' = C_r / V$ (Amp-Hr)
 - P_e : Power during Eclipse
 - T_e : Eclipse Duration
 - DOD: Depth of Discharge
 - N : Number of Batteries
 - n : Transmission Efficiency



◆ 11.4.2 Energy Storage (cont'd)

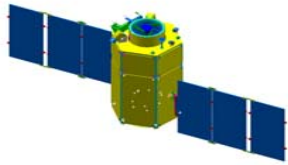
TABLE 11-40. Steps in the Energy Storage Subsystem Design. To obtain the required battery capacity in Amp-hr, divide by the required satellite bus voltage.

Step	Consider	FireSat Example
1. Determine the energy storage requirements	• Mission length • Primary or secondary power storage • Orbital parameters – Eclipse frequency – Eclipse length • Power use profile – Voltage and current – Depth of discharge – Duty cycles • Battery charge/discharge cycle limits	• 5 yrs • Secondary power storage • 16 eclipses per day • 35.3 min per eclipse (T_e) • Eclipse load 110 W (P_e) – 26.4 V, 4.2 A (max) • 20% (upper limit) • TBD—depends on observations taken and downlinked during eclipses
2. Select the type of secondary batteries	• NiCd (space qualified) • NiH₂ (space qualified) • Li-ion (under development) • NaS (under development)	• NiCd or NiH₂—both are space-qualified and have adequate characteristics
3. Determine the size of the batteries (battery capacity)	• Number of batteries • Transmission efficiency between the battery and the load	• $N = 3$ batteries (nonredundant) • $n = 0.90$ • $C_r = 119$ W-hr • $C_r = 4.5$ Amp-hr (26.4 V bus)
Battery Capacity: $C_r = \frac{P_e T_e}{(DOD) N n}$ W-hr (for battery capacity in Amp-hr, divide by bus voltage)		



◆ 11.4.3 Power Distribution

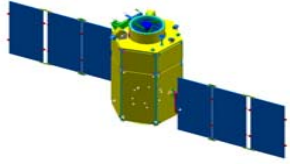
- S/C Power Distribution Subsystem consists of:
 - Cabling
 - Fault Protection
 - Switching Gear to turn power on and off to the S/C loads
- Systems for distributing power on S/C have been *DC* because S/C generate direct current power
- Harness or Cabling is a large part (10~25%) of EPS
- Centralized System
 - Regulates power to all S/C loads within the main bus
- Decentralized System
 - Places the converters at each load separately.



◆ 11.4.3 Power Distribution (cont'd)

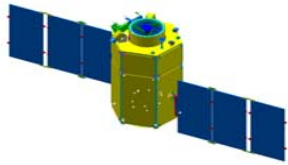
TABLE 11-41. Steps in the Power Distribution Subsystem Design.

Step	Consider	Possibilities
1. Determine the electrical load profile	• All spacecraft loads, their duty cycles, and special operating modes • Inverters for ac requirements • Transient behavior within each load • Load-failure isolation	• Low-voltage dc: 5 V • High-voltage dc: 270 V • High-voltage 1-phase ac: 115 V_{rms}, 60 Hz • High-voltage, 3-phase ac: 120/440 V_{rms}, 400 Hz
2. Decide on centralized or decentralized control	• Individual load requirements • Total system mass	• Converters at each load—for a few special loads • Centralized converters control voltage from the main bus (no specialized power requirements)
3. Determine the fault protection subsystem	• Detection (active or passive) • Isolation • Correction (change devices, reset fuses, work around lost subsystem)	• Cable size (length and diameter) and excess current-carrying ability • Size of power storage in case of a short circuit • Location of fuses and their type



◆ 11.4.4 Power Regulation and Control

- Power Regulation divided into three main categories:
 - Controlling the solar array
 - Must control electrical power generated at the array to prevent battery overcharging and undesired S/C heating
 - Peak–Power Tracking (PPT) Systems is a DC–DC converter operating in series with the solar array
 - Advantages for missions under 5 years that require more power at BOL than at EOL
 - Direct–Energy–Transfer (DET) System is a shunt regulation that operates in parallel to the array
 - Advantage are fewer plants, lower mass, higher total efficiency at EOL
 - Regulating bus voltage
 - Unregulated/Quasi–Regulated/Fully–Regulated System
 - Charging the battery
 - Parallel Charging (<5yrs): Simpler, Lower cost
 - Individual Charging (>5yrs)



◆ 11.4.4 Power Regulation and Control (cont'd)

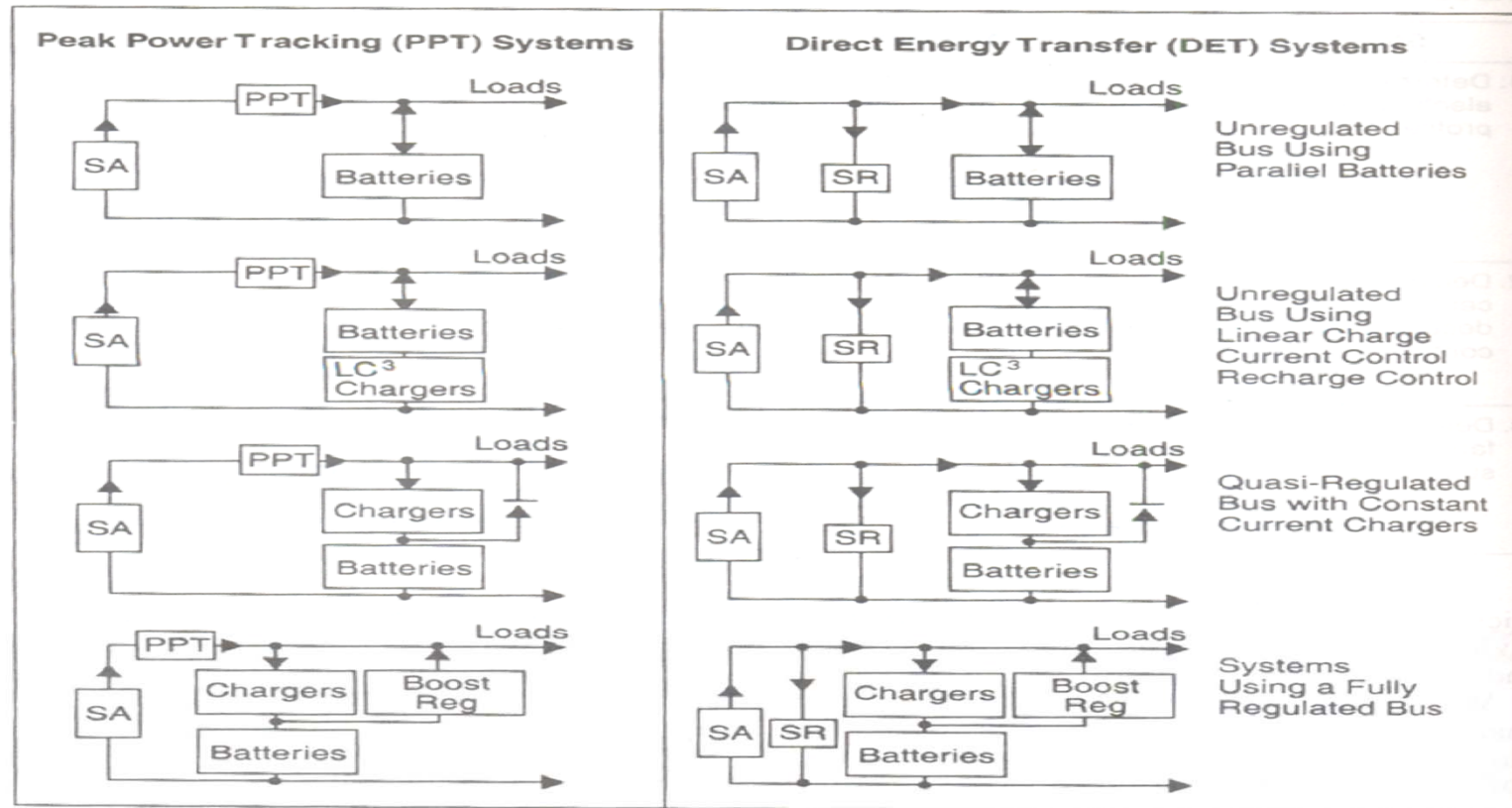
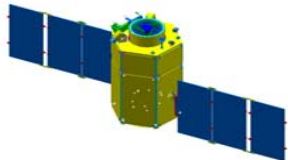


Fig. 11-13. Techniques for Power Regulation. The basic approaches are *Peak Power Tracking (PPT)*, which places a regulator in series with the solar arrays and the load, and *Direct Energy Transfer (DET)*, which uses a regulator in parallel with the solar arrays and load.

- SA: Solar Array
- SR: Shunt Regulator



◆ 11.4.4 Power Regulation and Control (cont'd)

TABLE 11-42. Steps in the Power Regulation and Control Subsystem Design.

Step	Consider	Possibilities
1. Determine the power source	• All spacecraft loads, their duty cycles, and special operating modes	• Primary batteries • Photovoltaic • Static power • Dynamic power
2. Design the electrical control subsystem	• Power source • Battery charging • Spacecraft heating	• Peak-power tracker • Direct-energy transfer
3. Develop the electrical bus voltage control	• How much control does each load require? • Battery voltage variation from charge to discharge • Battery recharge subsystem • Battery cycle life • Total system mass	• Unregulated • Quasi-regulated • Fully regulated • Parallel or individual charging – < 5 yrs—parallel charge – > 5 yrs—independent charge